

CREDIT CARD FRAUD DETECTION USING MACHINE LEARNING

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Python Libraries used:

- Numpy
- Pandas
- Matplotlib
- Scikit-learn

Project Objective

This project aims to build and compare multiple machine learning models for detecting fraudulent credit card transactions. Because fraudulent transactions are extremely rare, the project focuses on evaluating models using Precision, Recall and F1-score rather than relying only on accuracy.

Dataset

The dataset contains 284,807 credit card transactions, including 492 fraudulent transactions. It consists of 30 input features (Time, Amount and anonymized PCA-transformed features V1-V28) and one target variable (Class).

Data Preprocessing

After loading the dataset, missing values were checked and none were found. The target variable (Class) was separated from the input features. The Time and Amount columns were standardized using StandardScaler, while the PCA-transformed features were retained as provided. The dataset was split into training (80%) and testing (20%) sets using stratified sampling to preserve the class distribution.

Exploratory Data Analysis

Exploratory analysis included dataset inspection, summary statistics, class distribution and transaction amount analysis. The dataset was found to be highly imbalanced, with approximately 99.83% genuine transactions and only 0.17% fraudulent transactions. This highlighted the need to evaluate models using Precision, Recall and F1-score.

Model Development

Five machine learning algorithms were trained and compared: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM) and K-Nearest Neighbors (KNN). Each model was evaluated on the unseen test dataset.

Model Evaluation

Random Forest achieved the best overall performance among all baseline models. Although all models reported very high accuracy because of the class imbalance, F1-score was used as the primary evaluation metric to compare their ability to detect fraudulent transactions.

Model	Accuracy	Precision	Recall	F1 Score
Random Forest	0.999596	0.941176	0.816327	0.874317
KNN	0.999544	0.918605	0.806122	0.858696
SVM	0.999403	0.944444	0.693878	0.800000
Decision Tree	0.999140	0.752577	0.744898	0.748718
Logistic Regression	0.999157	0.828947	0.642857	0.724138

Feature Importance and Selection

Feature importance obtained from the Random Forest model was used to rank all input variables. The top 12 most important features were selected, and all models were retrained using only these features. This reduced the dimensionality of the dataset while maintaining or improving predictive performance.

Final Results

Using the selected features, Random Forest achieved the best overall performance:

Accuracy: 99.9614%

Precision: 94.19%

Recall: 82.65%

F1-score: 88.04%

The reduced-feature model slightly outperformed the model trained using all available features.

Model	Accuracy	Precision	Recall	F1 Score
Random Forest	0.999614	0.941860	0.826531	0.880435
KNN	0.999561	0.910112	0.826531	0.866310
SVM	0.999473	0.959459	0.724490	0.825581
Decision Tree	0.999192	0.760000	0.775510	0.767677
Logistic Regression	0.999105	0.813333	0.622449	0.705202

AFTER FEATURE SELECTION

Conclusion

This project demonstrated a complete machine learning workflow for an imbalanced classification problem. Multiple algorithms were compared, important features were identified using Random Forest, and feature selection improved the performance of several models while simplifying the input space. The project provides practical experience in data preprocessing, model evaluation and feature engineering for fraud detection.